

AI Overview



Objectives

Upon completion of this course, you will be able to:

- Understand basic concepts of AI.
- Understand the application technologies and application fields of AI.
- Know the development trends of AI.

Contents

- 1. AI Overview**
2. Technical Fields and Application Fields of AI
3. AI Disputes
4. Future Prospects of AI

AI in the Eyes of the Society

- People get to know AI through news, movies, and actual applications in daily life. What is AI in the eyes **of the public?**

Haidian Park: First AI-themed Park in the World
StarCraft II: AlphaStar Beat Professional Players
AI-created Edmond de Belamy Sold at US\$430,000
Demand for AI Programmers: ↑ 35 Times! Salary: Top 1!
50% Jobs Will be Replaced by AI in the future
Winter is Coming? AI Faces Challenges

...

News

AI Applications
AI industry outlook
Challenges faced by AI

...

The Terminator
2001: A Space Odyssey
The Matrix
I, Robot
Blade Runner
Elle
Bicentennial Man

...

Movies

AI Control over human beings
Fall in love with AI
Self-awareness of AI

...

Self-service security check
Spoken language evaluation
Music/Movie recommendation
Smart speaker
AI facial fortune-telling
Vacuum cleaning robot
Self-service bank terminal
Intelligent customer service
Siri

...

Applications in daily life

Security protection
Entertainment
Smart Home
Finance

...

AI in the Eyes of Researchers

"I propose to consider the question, 'Can machines think?'"

– Alan Turing 1950

The branch of computer science concerned with making computers behave like humans.

– John McCarthy 1956

The science of making machines do things that would require intelligence if done by men.

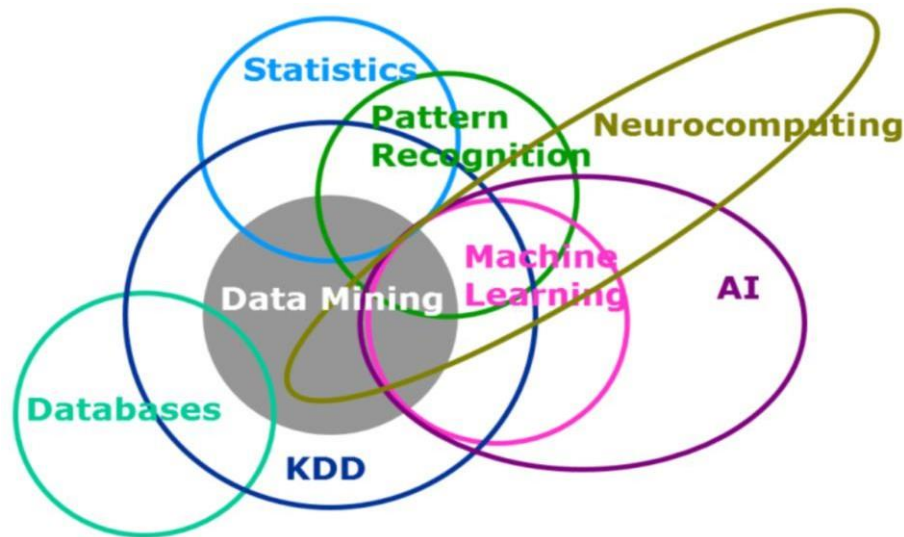
– Marvin Minsky

What Are Intelligences?

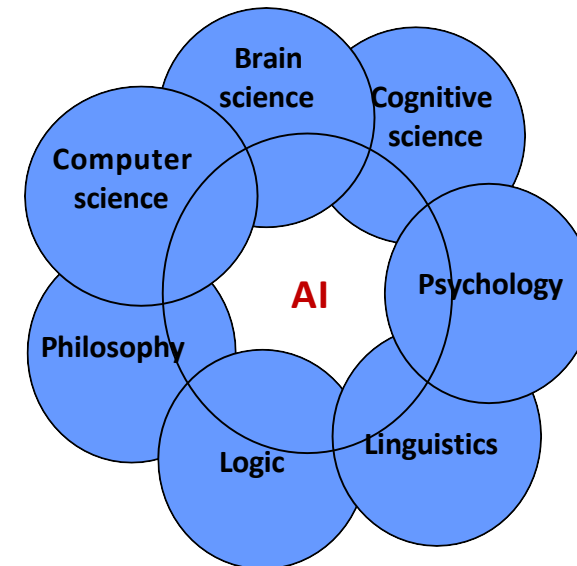
- Howard Gardner's Multiple Intelligences
- Human intelligences can be divided into seven categories:
 - Verbal/Linguistic
 - Logical/Mathematical
 - Visual/Spatial
 - Bodily/Kinesthetic
 - Musical/Rhythmic
 - Inter-personal/Social
 - Intra-personal/Introspective

What Is AI?

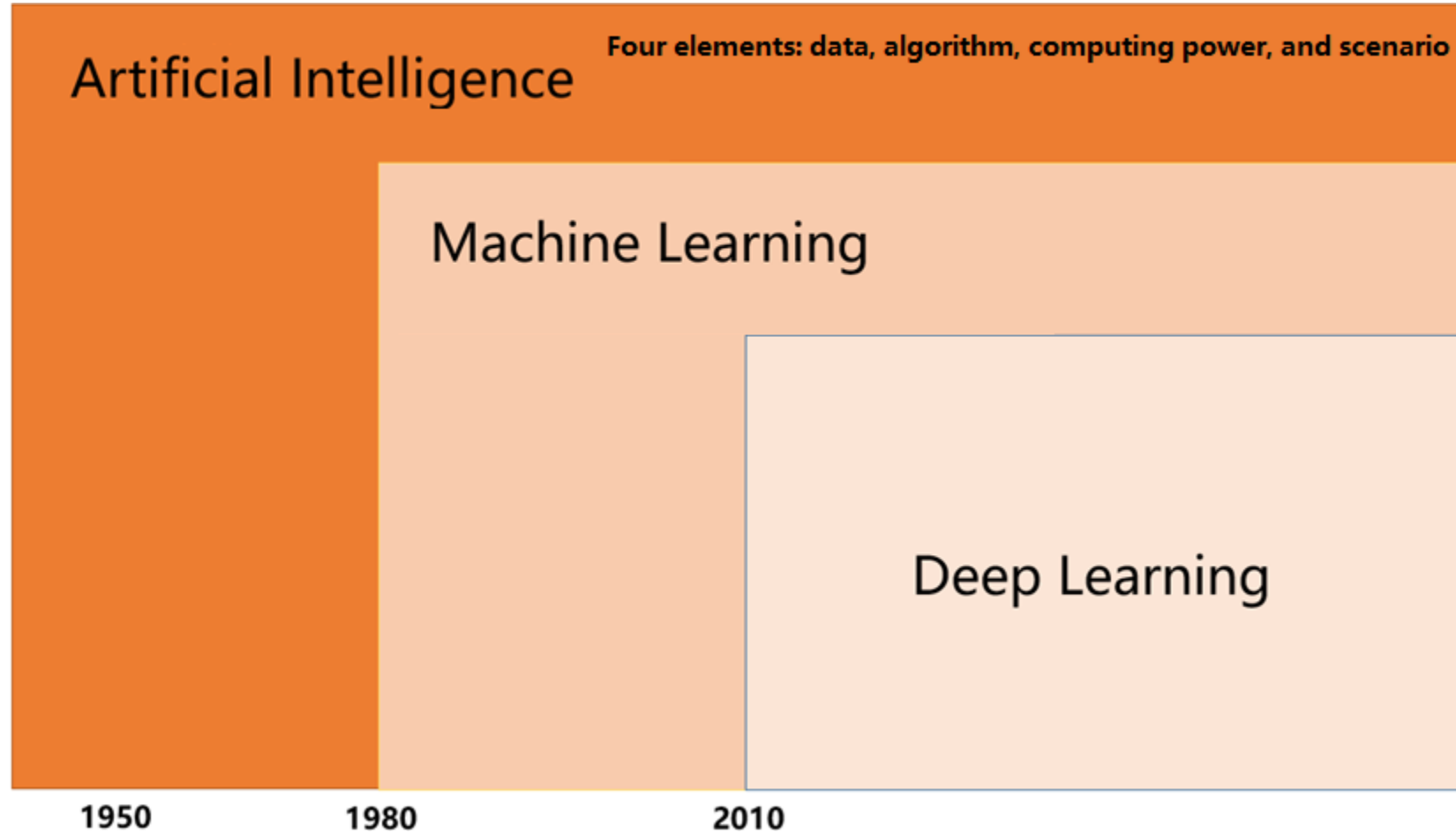
- Artificial Intelligence (AI) is a new technical science that studies and develops theories, methods, techniques, and application systems for simulating and extending human intelligence. In 1956, the concept of AI was first proposed by John McCarthy, who defined the subject as "science and engineering of making intelligent machines, especially intelligent computer program". AI is concerned with making machines work in an intelligent way, similar to the way that the human mind works. At present, AI has become an interdisciplinary course that involves various fields.



Identification of concepts related to AI and machine learning
AI Development Report 2020



Relationship of AI, Machine Learning, and Deep Learning

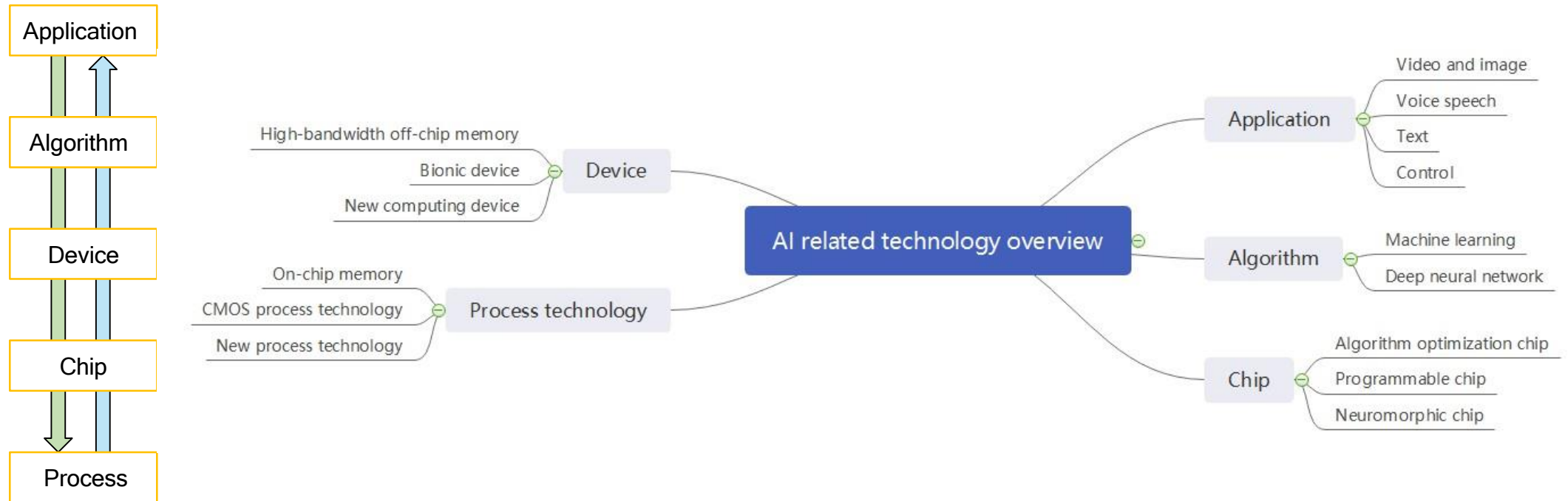


Relationship of AI, Machine Learning and Deep Learning

- AI: A new technical science that focuses on the research and development of theories, methods, techniques, and application systems for simulating and extending human intelligence.
- Machine learning: A core research field of AI. It focuses on the study of how computers can obtain new knowledge or skills by simulating or performing learning behavior of human beings, and reorganize existing knowledge architecture to improve its performance. It is one of the core research fields of AI.
- Deep learning: A new field of machine learning. The concept of deep learning originates from the research on artificial neural networks. The multi-layer perceptron (MLP) is a type a deep learning architecture. Deep learning aims to simulate the human brain to interpret data such as images, sounds, and texts.

Overview of AI Technologies

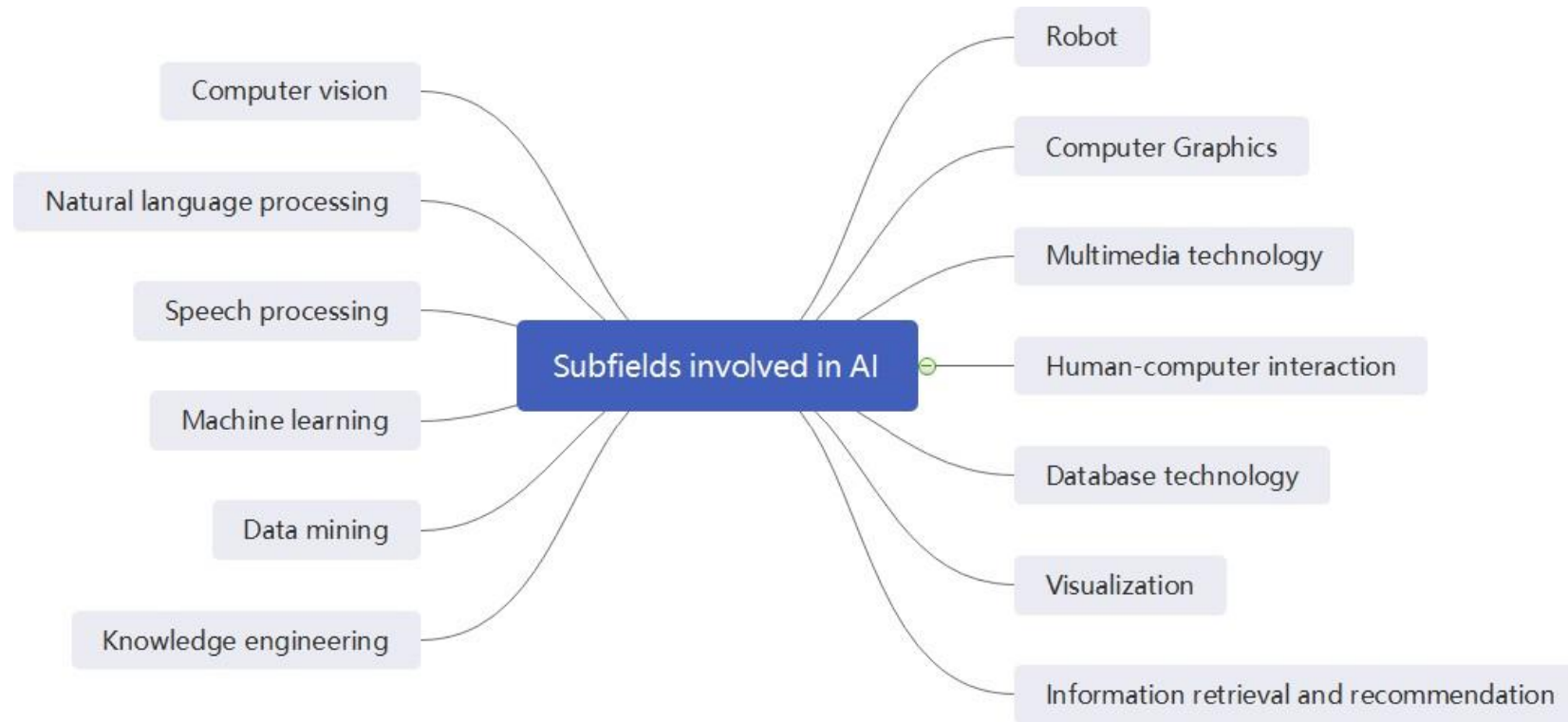
- AI technologies are multi-layered, covering the application, algorithm mechanism, toolchain, device, chip, process, and material layers.



Types of AI

- Strong AI
 - The strong AI view holds that it is possible to create intelligent machines that can really reason and solve problems. Such machines are considered to be conscious and self-aware, can independently think about problems and work out optimal solutions to problems, have their own system of values and world views, and have all the same instincts as living things, such as survival and security needs. It can be regarded as a new civilization in a certain sense.
- Weak AI
 - The weak AI view holds that intelligent machines cannot really reason and solve problems. These machines only look intelligent, but do not have real intelligence or self-awareness.

Sub-fields of AI

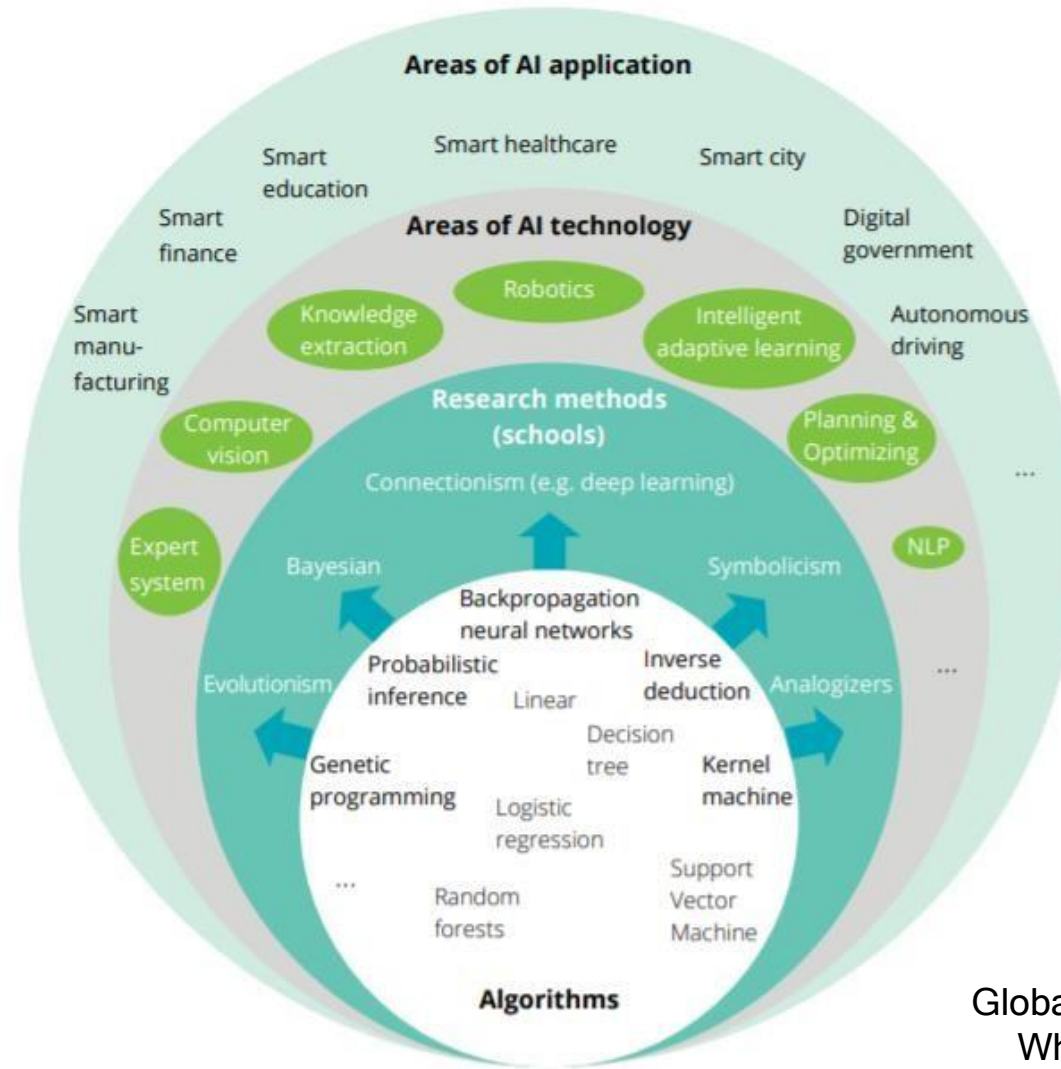


AI Development Report 2020

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- 2. Technical Fields and Application Fields of AI**
3. AI Disputes
4. Future Prospects of AI

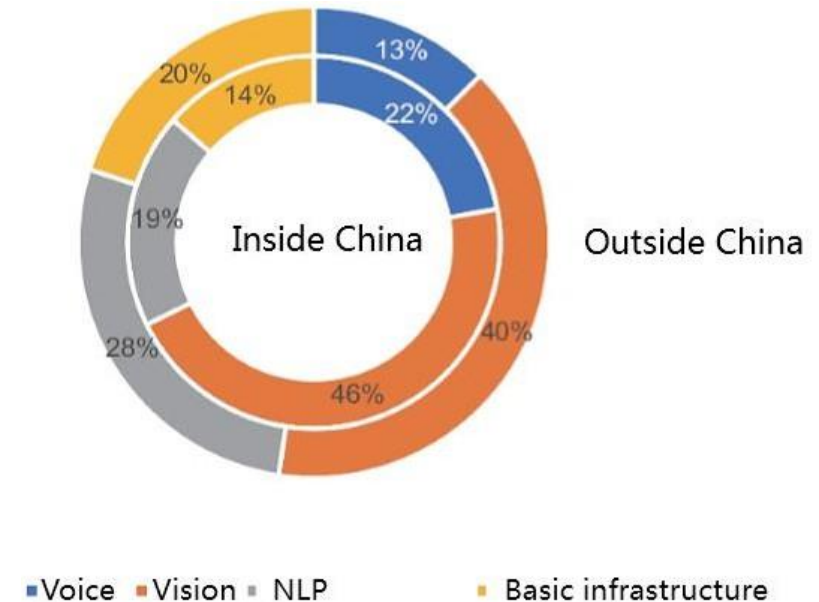
Technical Fields and Application Fields of AI



Global AI Development
White Paper 2020

Distribution of AI Application Technologies in Enterprises

- At present, application directions of AI technologies mainly include:
 - **Computer vision:** a science of how to make computers "see"
 - **Speech processing:** a general term for various processing technologies used to research the voicing process, statistical features of speech signals, speech recognition, machine-based speech synthesis, and speech perception
 - **Natural language processing (NLP):** a subject that use computer technologies to understand and use natural language



Distribution of AI application technologies in enterprises inside and outside China

China AI Development Report 2018

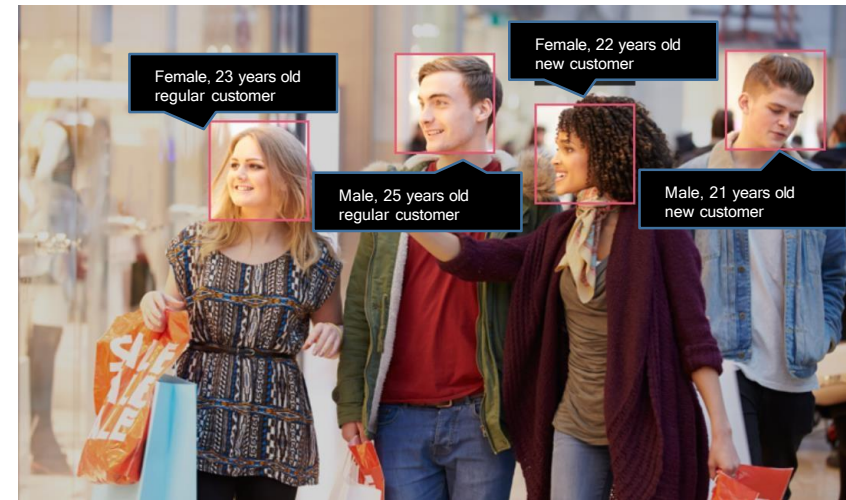
Computer Vision Application Scenario (1)

- Computer vision is the most mature technology among the three AI technologies. The main topics of computer vision research include image classification, target detection, image segmentation, target tracking, optical character recognition (OCR), and facial recognition.
- In the future, computer vision is expected to enter the advanced stage of autonomous understanding, analysis, and decision-making, enabling machines to "see" and bringing greater value to scenarios such as unmanned vehicles and smart homes.
- Application scenarios:

Facial recognition Comparison Gallery Authentication result



Electronic attendance



Traffic analysis

Computer Vision Application Scenario (2)

Action analysis



Authentication



Smart album

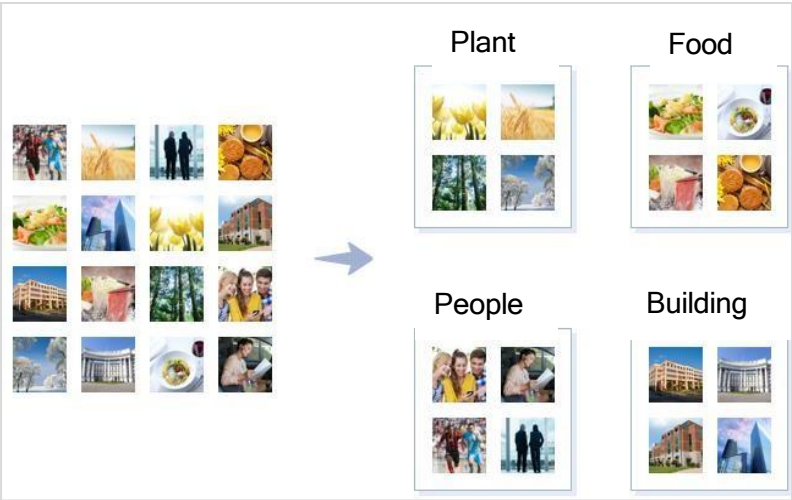
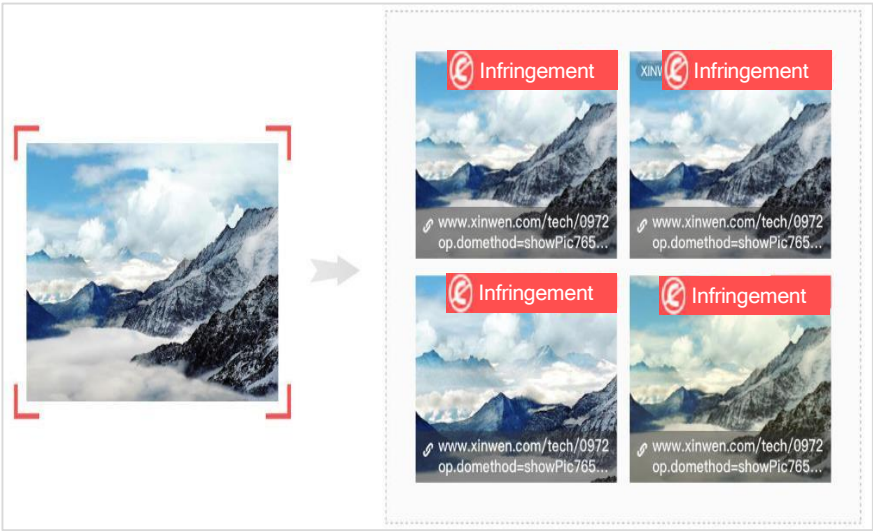


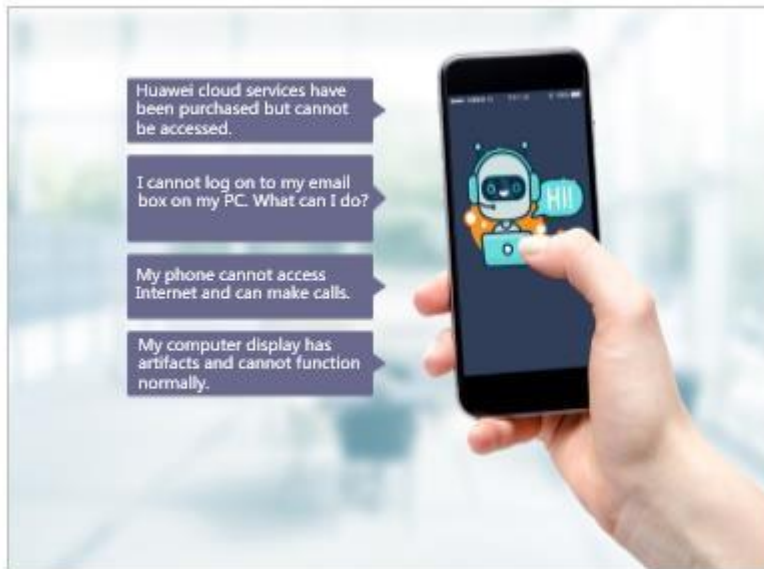
Image search



Voice Processing Application Scenario (1)

- The main topics of voice processing research include voice recognition, voice synthesis, voice wakeup, voiceprint recognition, and audio-based incident detection. Among them, the most mature technology is voice recognition. As for near field recognition in a quite indoor environment, the recognition accuracy can reach 96%.
- Application scenarios:

Question Answering Bot (QABot)



Voice navigation



Voice Processing Application Scenario (2)

Intelligent education



The Chinese phonetic alphabet consists of initials and finals... The Chinese tones include...

Real-time conference records



- Other applications:
 - Spoken language evaluation
 - Diagnostic robot
 - Voiceprint recognition
 - Smart sound box
 - ...

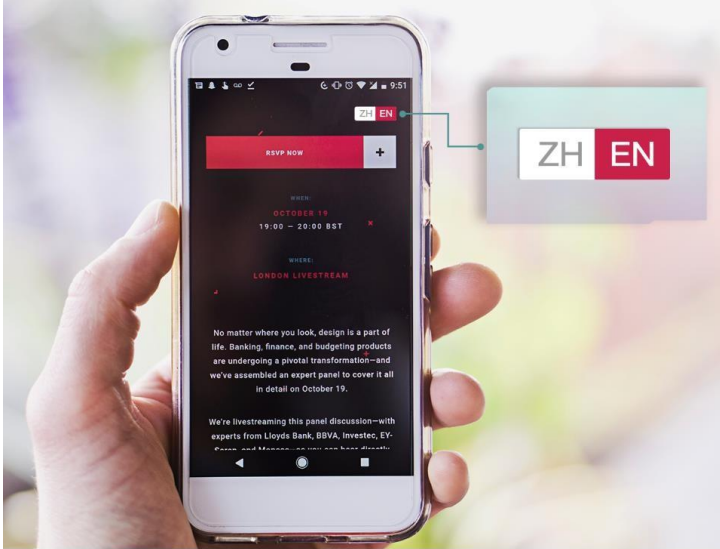
NLP Application Scenario (1)

- The main topics of NLP research include machine translation, text mining, and sentiment analysis. NLP imposes high requirements on technologies but confronts low technology maturity. Due to high complexity of semantics, it is hard to reach the human understanding level using parallel computing based on big data and parallel computing only.
- In future, NLP will achieve more growth: understanding of shallow semantics → automatic extraction of features and understanding of deep semantics; single-purpose intelligence (ML) → hybrid intelligence (ML, DL, and RL)
- Application scenarios:



NLP Application Scenario (2)

Machine translation



Text classification



- Other applications:
 - Knowledge graph
 - Intelligent copywriting
 - Video subtitle
 - ...

AI Application Field - Intelligent Healthcare

Medicine mining: quick development of personalized medicines by AI assistants

Health management: nutrition, and physical/mental health management **Hospital**

management: structured services concerning medical records (focus) **Assistance for**

medical research: assistance for biomedical researchers in research

Virtual assistant: electronic voice medical records, intelligent guidance, intelligent diagnosis, and medicine recommendation

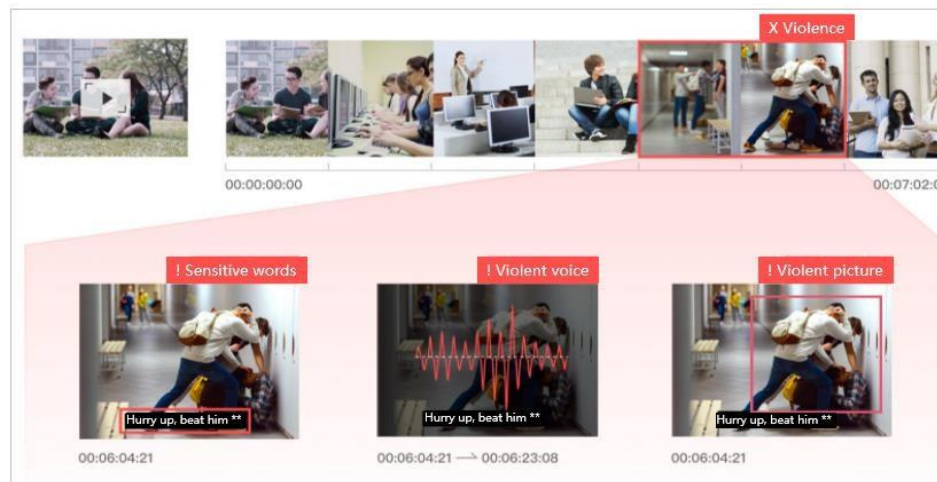
Medical image: medical image recognition, image marking, and 3D image reconstruction

Assistance for diagnosis and treatment: diagnostic robot

Disease risk forecast: disease risk forecast based on gene sequencing

AI Application Field - Intelligent Security

- Security protection is considered the easiest field for AI implementation. AI technologies applied in this field are relatively mature. The field involves massive data of images and videos, laying a sound foundation for training of AI algorithms and models. Currently, AI technologies are applied to two directions in the security protection field, namely, civil use and police use.
- Application scenarios:
 - **Police use:** suspect identification, vehicle analysis, suspect tracking, suspect search and comparison, and access control at key places
 - **Civil use:** facial recognition, warning against potential danger, and home protective measure deployment



AI Application Field - Smart Home

- Based on IoT technologies, a smart home ecosystem is formed with hardware, software, and cloud platforms, providing users personalized life services and making home life more convenient, comfortable, and safe.

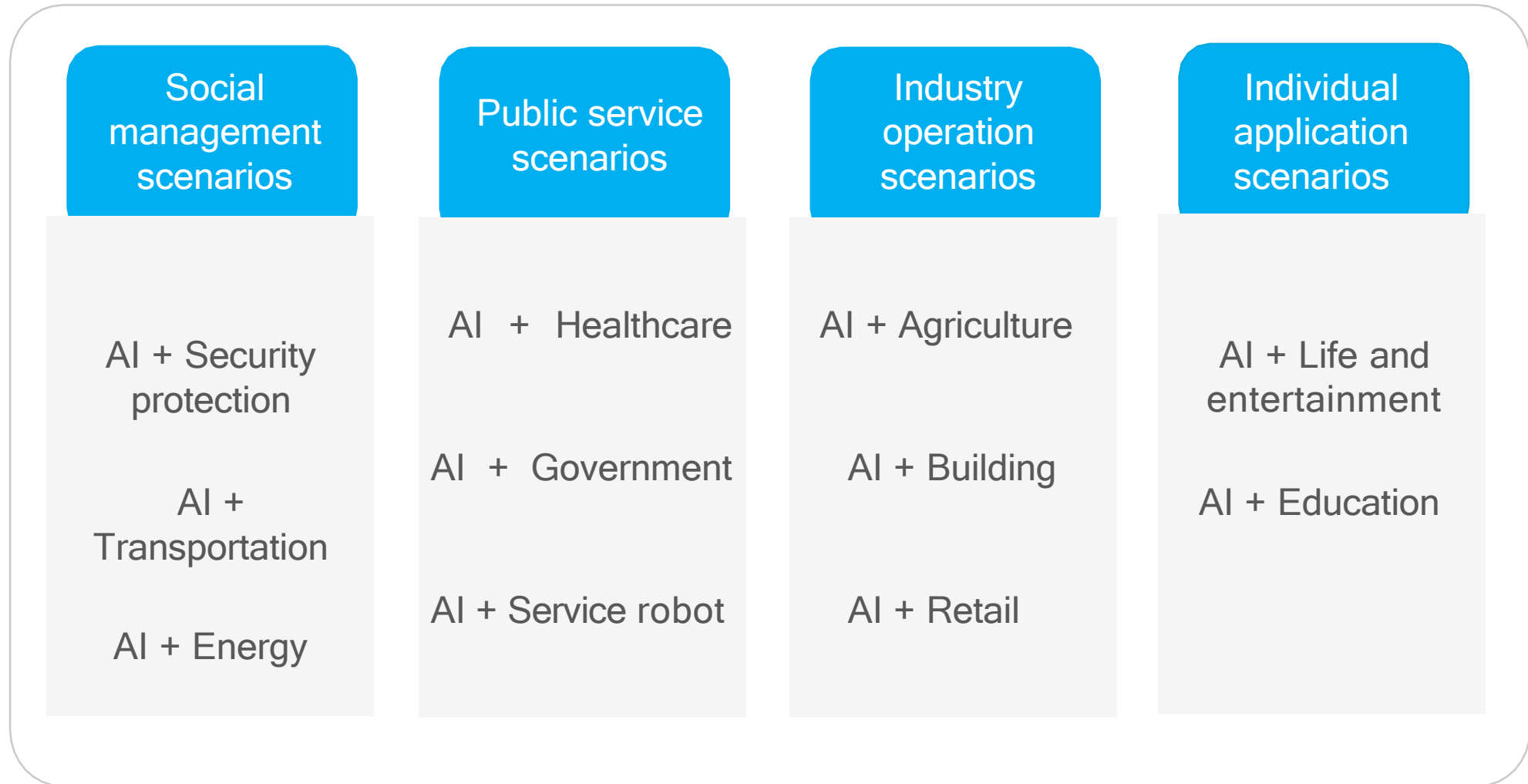
Control smart home products with voice processing such as air conditioning temperature adjustment, curtain switch control, and voice control on the lighting system.

Implement home security protection with computer vision technologies, for example, facial or fingerprint recognition for unlocking, real-time intelligent camera monitoring, and illegal intrusion detection.

Develop user profiles and recommend content to users with the help of machine learning and deep learning technologies and based on historical records of smart speakers and smart TVs.



AI Application Field - Smart City



AI Application Field - Retail

- AI will bring revolutionary changes to the retail industry. A typical symptom is unmanned supermarkets. For example, Amazon Go, unmanned supermarket of Amazon, uses sensors, cameras, computer vision, and deep learning algorithms to completely cancel the checkout process, allowing customers to pick up goods and "just walk out".
- One of the biggest challenges for unmanned supermarket is how to charge the right fees to the right customers. So far, Amazon Go is the only successful business case and even this case involves many controlled factors. For example, only Prime members can enter Amazon Go. Other enterprises, to follow the example of Amazon, have to build their membership system first.



AI Application Field - Autonomous Driving

- The Society of Automotive Engineers (SAE) in the U.S. defines 6 levels of driving automation ranging from 0 (fully manual) to 5 (fully autonomous). L0 indicates that the driving of a vehicle completely depends on the driver's operation. The system above L3 can implement the driver's hand-off operation in specific cases, L5 depends on the system when vehicles are driving in all scenarios.
- Currently, only some commercial passenger vehicle models, such as Audi A8, Tesla, and Cadillac, support L2 and L3 Advanced driver-assistance systems (ADAS). It is estimated that by 2020, more L3 vehicle models will emerge with the further improvement of sensors and vehicle-mounted processors. L4 and L5 autonomous driving is expected to be first implemented on **commercial vehicles in closed campuses**. A wider range of passenger vehicles require advanced autonomous driving, which requires further improvement of technologies, policies, and infrastructure. It is estimated that L4 and L5 autonomous driving will be supported by common roads in 2025-2030.

AI Will Change All Industries



Public sector

- Safe City
- Intelligent transport
- Disaster prediction



Education

- Personalization
- Attention improvement
- Robot teacher



Healthcare

- Early prevention
- Diagnosis assistance
- Precision cure



Media

- Real-time translation
- Abstraction
- Inspection



Pharmacy

- Fast R&D
- Precise trial
- Targeted medicine



Logistics

- Routing planning
- Monitoring
- Auto sorting



Finance

- Doc process
- Real-time fraud prevention
- Up-sell



Insurance

- Auto detection
- Fraud prevention
- Innovative service



Retail

- Staff-less shops
- Real-time inventory
- Precise recommendations



Manufacturing

- Defect detection
- Industrial internet
- Predictive maintenance



Telecom

- Customer service
- Auto O&M
- Auto optimization



Oil and gas

- Localization
- Remote maintenance
- Operation optimization



Agriculture

- Fertilization improvement
- Remote operation
- Seeds development

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Algorithmic Bias

- Algorithmic biases are mainly caused by data biases.
- When we use AI algorithms for decision-making, the algorithms may learn to discriminate an individual based on existing data including race and gender, and therefore create unfair outcomes, such as decisions that are discriminatory based on race, sex or other factors. Even if factors such as race or gender are excluded from the data, the algorithms can make discriminatory decisions based on information of names and addresses.

If we search with a name sounds like an African American, an advertisement for **a tool used to search criminal records** may be displayed. The advertisement, however, is not likely displayed in other cases.

Online advertisers tend to display advertisements of lower-priced goods to **female users**.

Google's image software once mistakenly labeled an image of black people as **"gorilla"**.

Privacy Issues

- The existing AI algorithms are all data-driven. In this case, we need a large amount of data to train models. We enjoy the convenience brought by AI every day while technology companies like Facebook, Google, Amazon, and Alibaba are obtaining an enormous amount of user data, which will reveal various aspects of our lives including politics, religions, and gender.

In principle, technology companies can record each click, each page scrolling, time of viewing any content, and browsing history when users access the Internet.

Technology companies can know our privacy including where are we, where we go, what we have done, education background, consumption capabilities, and personal preferences based on our ride-hailing records and consumption records.

Seeing = Believing?

- With the development of computer vision technologies, reliability of images and videos is decreasing. Fake images can be produced with technologies such as PS and generative adversarial networks (GAN), making it hard to identify whether images are true or not.
- Example:
 - A suspect provided fake evidence by forging an image in which the suspect is in a place where he has never been to or with someone he has never seen using PS technologies.
 - In advertisements for diet pills, people's appearances before and after weight loss can be changed with PS technologies to exaggerate the effect of the pills.
 - Lyrebird, a tool for simulating voice of human beings based on recording samples of minutes, may be used by criminals.
 - Household images released on rent and hotel booking platforms may be generated through GAN.

AI Development = Rising Unemployment?

- Looking back, human beings have always been seeking ways to improve efficiency, that is, obtain more with less resources. We used sharp stones to hunt and collect food more efficiently. We used steam engines to reduce the need for horses. Every step in achieving automation will change our life and work. In the era of AI, what jobs will be replaced by AI?
- The answer is repetitive jobs that involve little creativity and social interaction.

Jobs Most Likely to Be Replaced by AI	Jobs Most Unlikely to Be Replaced by AI
Courier	Writer
Taxi driver	Management personnel
Soldier	Software engineers
Accounting	HR manager
Telesales personnel	Designer
Customer service	Activity planner
...	...

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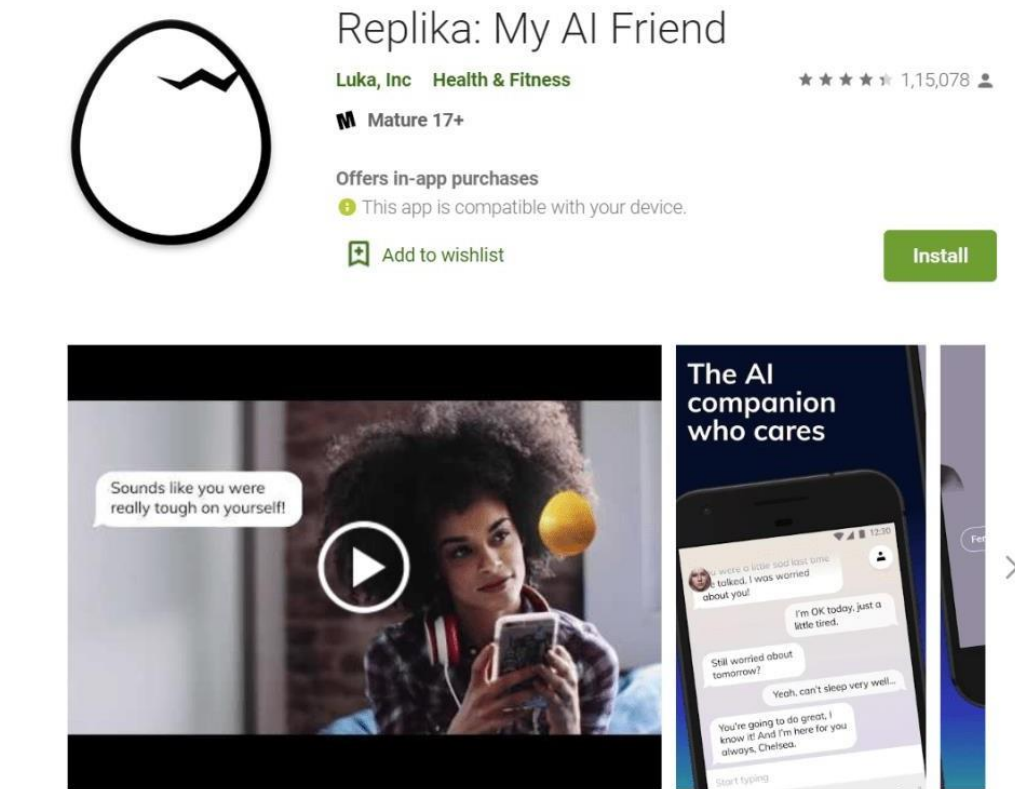
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Development Trends of AI Technologies

- Framework: easier-to-use development framework
- Algorithm: algorithm models with better performance and smaller size
- Computing power: comprehensive development of device-edge-cloud computing
- Data: more comprehensive basic data service industry and more secure data sharing
- Scenario: continuous breakthroughs in industry applications

Mitigating Psychological Problems

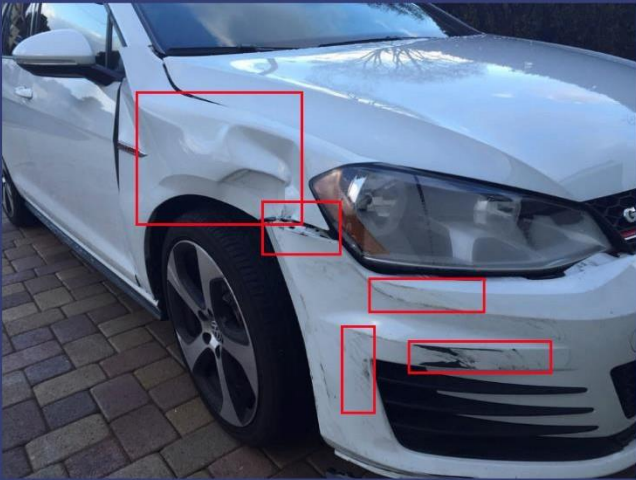
- AI chat robots help alleviate mental health problems such as autism by combining psychological knowledge.



Automatic Vehicle Insurance and Loss Assessment

- AI technologies help insurance companies optimize vehicle insurance claims and complete vehicle insurance loss assessment using deep learning algorithms such as image recognition.

Vehicle Damage Assessment



```
{
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    {
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      "score": "high",
      "xmin": 210,
      "ymin": 139,
      "xmax": 474,
      "ymax": 351
    },
    {
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      "score": "high",
      "xmin": 408,
      "ymin": 312,
      "xmax": 535,
      "ymax": 399
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      "score": "medium",
      "xmin": 639,
      "ymin": 532,
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      "ymax": 585
    }
  ]
}
```

Office Automation

- AI is automating management, but the different nature and format of data makes it a challenging task. While each industry and application has its own unique challenges, different industries are gradually adopting machine learning-based workflow solutions.



Thank you